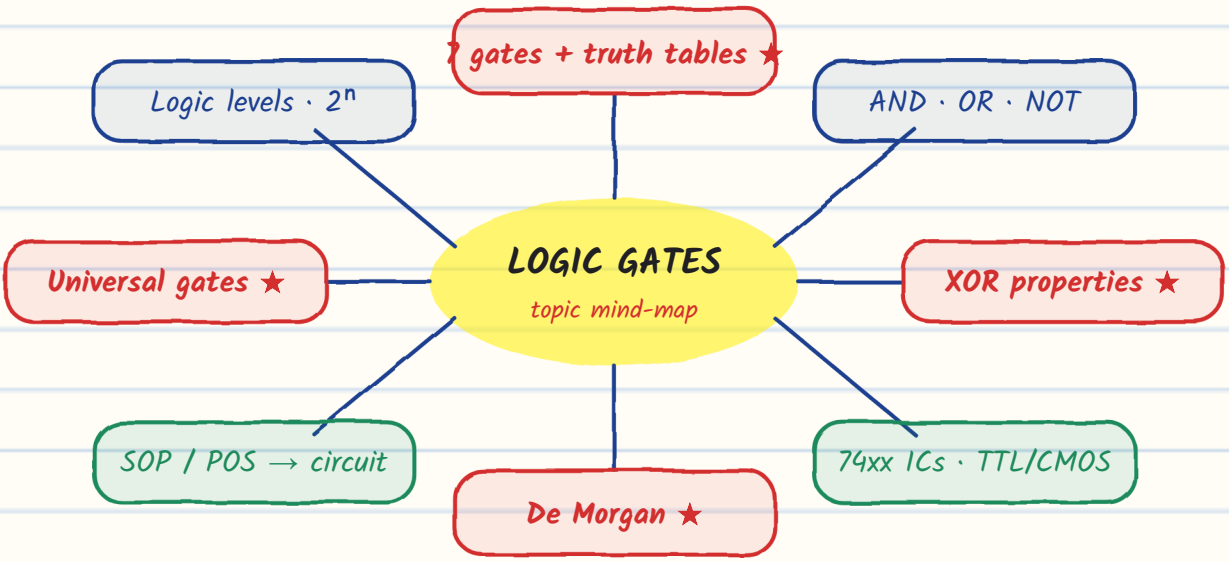
Logic Gates

Truth tables · universal gates · De Morgan · ICs: exam-ready in 10 sheets



↑ the whole topic on one page: red = high-yield, green = applied

➜ Syllabus coverage

"Digital logic: Boolean algebra, De Morgan's Theorems, logic gates and their truth tables, canonical forms, combinational logic circuits, minimization techniques". This notebook covers the gate part in depth and bridges into canonical forms; K-maps are in the Boolean Algebra lesson.

High-yield list

① 2-input truth tables
② min. NAND/NOR counts
③ De Morgan & bubbles
④ XOR = odd parity
⑤ 7400/02/04/08/32/86
⑥ +ve vs −ve logic

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

➜ Contents

✓ Fundamentals & logic levels ..... p. 2
✓ AND, OR, NOT ..... p. 3
✓ NAND, NOR, XOR, XNOR ..... p. 4
✓ Universal gates & counts ..... p. 5
✓ De Morgan & bubble pushing ..... p. 6
✓ Expression → circuit (SOP/POS) ... p. 7
✓ ICs, families, applications ..... p. 8
✓ Night-before sheet ..... p. 9
✓ Practice MCQs + written ..... p. 10

1 What Is a Logic Gate? MCQ + Written

Definition A logic gate is an electronic circuit that takes one or more binary inputs and produces one binary output according to a fixed Boolean function. Gates are the building blocks of every digital system: ALU, memory, CPU control.

Boolean operation	Symbol	Gate	Output is 1 when ...
Logical product	A·B	AND	all inputs are 1
Logical sum	A+B	OR	at least one input is 1
Complement	A'	NOT	the input is 0

↙ Gate = Boolean operator made of transistors

Counting rules
n inputs → 2^n rows in the truth table
n variables → $2^{(2^n)}$ different Boolean functions
(n = 2 → 4 rows, **16** functions)

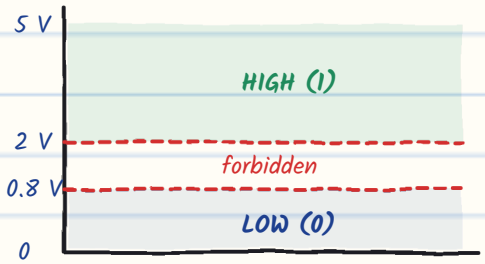
Memory trick Write the input columns like binary counting: 00, 01, 10, 11. You can never miss a row.

2 Logic Levels: Positive vs Negative Logic

A gate does not "see" 0 and 1; it sees **voltage ranges**.

- Positive logic: HIGH = 1, LOW = 0 (the default)
- Negative logic: HIGH = 0, LOW = 1

Same circuit, different labels → a different function.



TTL input thresholds

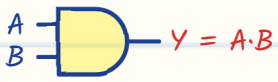
TTL (5 V) level	Input	Output
LOW (0)	$V_{IL} \le$ 0.8 V	$V_{OL} \le 0.4$ V
HIGH (1)	$V_{IH} \ge$ 2.0 V	$V_{OH} \ge 2.4$ V

Noise margin = 2.4 − 2.0 = 0.8 − 0.4 = **0.4 V**. CMOS thresholds scale with the supply (roughly 30% / 70% of V_{DD}).

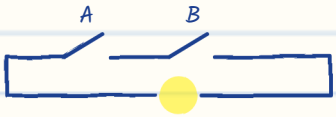
Exam trap A positive-logic **AND** gate behaves as an **OR** gate in negative logic; positive **NAND** = negative **NOR** (De Morgan in disguise).

3 The Basic Gates: AND, OR, NOT

→ AND: output 1 only if ALL inputs are 1 · Y = A·B

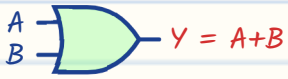


A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

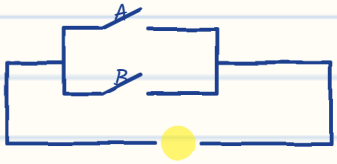


switches in series

→ OR: output 0 only if ALL inputs are 0 · Y = A + B



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



switches in parallel

→ NOT (inverter): output = complement · Y = A'



A	Y
0	1
1	0

The only **single-input** gate. The bubble ○ on any symbol means NOT. Two NOTs in a row cancel: (A')' = A.

→ Multi-input gates

- n-input AND: 1 only for 11...1 (one row out of 2^n)
- n-input OR: 0 only for 00...0
- AND, OR are associative: (A·B)·C = A·(B·C), so 2-input gates can be cascaded

Gating (enable / inhibit)

AND: control 1 → **passes** data, 0 → forces 0

OR: control 0 → passes data, 1 → forces 1

XOR: control 1 → **inverts** data

Memory trick **AND = "all", OR = "any".** AND looks for one 0 to kill the output; OR looks for one 1 to set it.

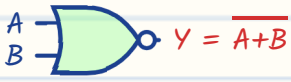
4 NAND, NOR, XOR, XNOR high-yield

NAND = NOT + AND



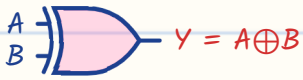
$Y = (AB)'$ · 0 **only** when all inputs are 1

NOR = NOT + OR



$Y = (A+B)'$ · 1 **only** when all inputs are 0

XOR (exclusive-OR)



$A \oplus B$ = $A'B + AB'$ · 1 when inputs **differ**

XNOR (equivalence)



$A \odot B$ = $AB + A'B'$ · 1 when inputs are **equal**

→ Master truth table (all 2-input gates)

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

XOR properties

↙ Complement pairs: AND↔NAND, OR↔NOR, XOR↔XNOR

$A \oplus 0 = A$ (pass) $A \oplus 1 = A'$ (invert)

$A \oplus A = 0$ $A \oplus A' = 1$

commutative & associative $(A \oplus B)' = A' \oplus B = A \oplus B'$

n-input XOR = 1 when the number of 1s is odd → **odd-parity function**; used for parity generators/checkers, comparators, adders, controlled inverters.

Exam trap NAND and NOR are **not associative**: (A NAND B) NAND C ≠ A NAND (B NAND C). A 3-input NAND is (ABC)', not a cascade of two 2-input NANDs.

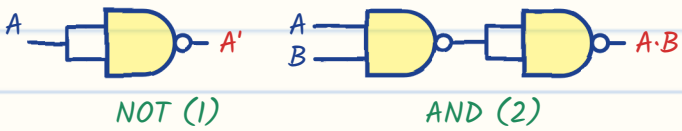
Memory trick "N" in front = put a bubble on the output = flip that gate's column. XOR = "**different** → 1", XNOR = "**same** → 1".

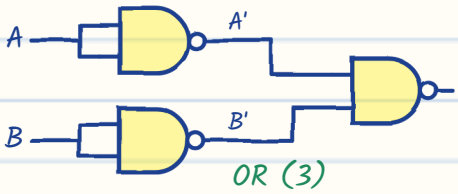
5 Universal Gates: NAND & NOR

written favourite

Definition A gate is *universal* if NOT, AND and OR (hence **any** Boolean function) can be built from it alone. **NAND** and **NOR** are universal; AND, OR, XOR are not.

→ Only NAND





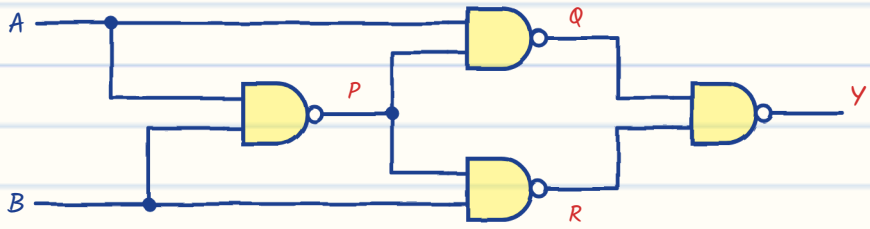
tie inputs together: NAND(A, A) = A'

Only NOR (mirror image): NOR(A, A) = A' (1 gate) · NOR then NOR-inverter = **OR** (2) · invert both inputs, then NOR: (A' + B')' = **AND** (3).

Function	NOT	AND	OR	XOR	XNOR
Min. 2-input **NAND** gates	1	2	3	4	5
Min. 2-input **NOR** gates	1	3	2	5	4

→ XOR with 4 NAND gates (the minimum)

↙ NAND and NOR swap the AND/OR and XOR/XNOR counts (duality)



① P = (AB)' = A' + B'
② Q = (A·P)' = (A(A'+B'))' = (AB')'
③ R = (B·P)' = (A'B)'
④ Y = (Q·R)' = AB' + A'B = **A ⊕ B**

Exam trap Many textbooks draw XOR with **5 NAND** or **6 NOR** (inverters + SOP). Those work but are not minimal; MCQs ask the **minimum: 4 NAND, 5 NOR.**

6 De Morgan's Theorems high-yield

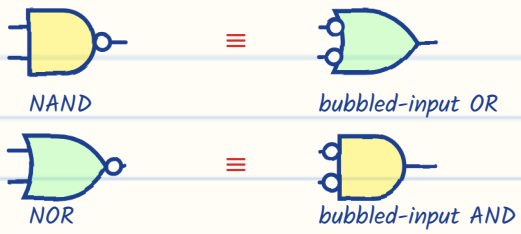
The two laws $(A \cdot B)' = A' + B'$ $(A + B)' = A' \cdot B'$

n variables: $(A \cdot B \cdot C \ldots)' = A' + B' + C' + \ldots$ · $(A + B + C + \ldots)' = A' \cdot B' \cdot C' \ldots$

→ Proof by truth table

A	B	(AB)'	A'+B'	(A+B)'	A'B'
0	0	1	1	1	1
0	1	1	1	0	0
1	0	1	1	0	0
1	1	0	0	0	0

→ Equivalent symbols



Memory trick "**Break the bar, change the sign**". In symbols: **bubble pushing**: move bubbles from output to all inputs (or back) and swap AND ↔ OR.

→ Worked example (written-answer format)

Q Simplify $F = (A + B)' + (A' + B')'$ and name the gate it equals. 4 marks

1. De Morgan on each term: $(A + B)' = A'B'$ and $(A' + B')' = A''B'' = AB$
2. So $F = A'B' + AB$
3. This is 1 when $A = B \rightarrow F = A \odot B$ (XNOR)
4. Circuit: one XNOR gate replaces 2 NOR + 2 NOT + 1 OR → fewer gates, lower cost & delay

7 Duality Principle

Rule Every Boolean identity stays true if you swap **+** ↔ **·** and **0** ↔ **1** (variables are **not** complemented).

Identity	Its dual
$A + 0 = A$	$A \cdot 1 = A$
$A + A' = 1$	$A \cdot A' = 0$
$A(B + C) = AB + AC$	$A + BC = (A + B)(A + C)$
$(A + B)' = A'B'$	$(AB)' = A' + B'$

Exam trap The dual is **not** the complement. Complement also flips every variable ($A \rightarrow A'$); the dual does not.

8 From Truth Table to Gate Circuit

Minterm (m) AND term with every variable, = 1 for exactly one row. **SOP** = OR of minterms = **Σm**

Maxterm (M) OR term with every variable, = 0 for exactly one row. **POS** = AND of maxterms = **ΠM**

1. Write the truth table (2^n rows)
2. Rows with F = 1 → minterms (SOP)
3. Simplify (Boolean laws or K-map)
4. Draw AND-OR (or NAND-NAND)
5. Check every row again

→ Worked example: 3-input majority voter

F = 1 when **at least two** of A, B, C are 1 (e.g. 2-out-of-3 approval).

#	A	B	C	F
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

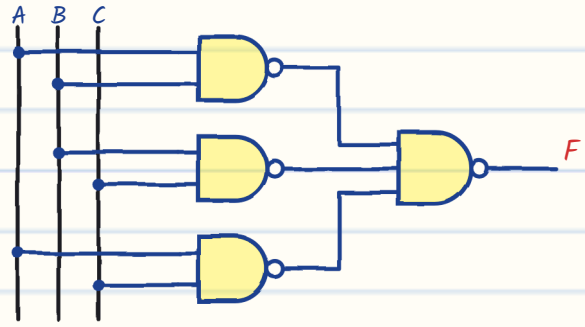
Canonical SOP: $F = \Sigma m(3, 5, 6, 7) = A'BC + AB'C + ABC' + ABC$

Canonical POS: $F = \Pi M(0, 1, 2, 4)$

Simplify (add ABC twice, A + A' = 1):

$$F = AB + BC + CA$$

= (A + B)(B + C)(C + A) in POS form



NAND-NAND: 3 two-input + 1 three-input NAND

Two-level rules

SOP (AND-OR) → replace every gate by **NAND** = NAND-NAND

POS (OR-AND) → replace every gate by **NOR** = NOR-NOR

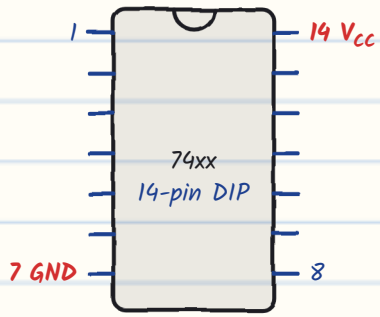
Why: F = ((AB)'(BC)'(CA)')' = AB + BC + CA by De Morgan

Fewer literals → fewer gates. Systematic minimisation (K-map, don't-cares) → see the **Boolean Algebra** lesson.

Exam trap The minterm list and the maxterm list are **complements**: together they cover all 2^n row numbers (here {3,5,6,7} ∪ {0,1,2,4}).

9 Gate ICs (74xx TTL family)

IC	Contents
7400	quad 2-input NAND
7402	quad 2-input NOR
7404	hex inverter (6 NOT)
7408	quad 2-input AND
7432	quad 2-input OR
7486	quad 2-input XOR
74266	quad 2-input XNOR (open collector)



notch up, pin 1 top-left, count anticlockwise

"Quad" = 4 gates, "hex" = 6 gates. Standard 14-pin DIP: V_{CC} = pin 14, GND = pin 7

10 Logic Families & Gate Parameters

	TTL (74xx)	CMOS (4000 / 74HC)
built from	bipolar transistors	MOSFETs (n + p pairs)
supply	5 V ± 5%	3–15 V (4000), 2–6 V (74HC)
static power	higher (mW per gate)	**very low** → used in CPUs, phones
fan-out	about 10 (standard TTL)	large (gate inputs draw ~no current)

- **Fan-in** = number of inputs of a gate · **Fan-out** = number of gate inputs one output can drive
- **Propagation delay** = input-to-output time (about 10 ns for standard TTL) → limits clock speed
- **Noise margin** = how much noise a level tolerates (TTL: 0.4 V)
- Integration scale (typical convention): **SSI** < 12 gates · **MSI** 12–99 · **LSI** 100–9,999 · **VLSI** ≥ 10,000

11 Where Gates Work

Classic circuits

Half adder: $S = A \oplus B$, $C = AB$

Even-parity bit for A, B, C: $P = A \oplus B \oplus C$

Equality (1-bit comparator): $A \odot B$

IT link: C bitwise

12 & 10 = **8** · 12 | 10 = **14** · 12 ^ 10 = **6** · ~12 = −13

Masks: x & 1 → odd/even test · x | mask sets bits · x ^ mask toggles

Swap: a ^= b; b ^= a; a ^= b;

Memory trick 74 + gate code: **00** NAND, **02** NOR, **04** NOT, **08** AND, **32** OR, **86** XOR. "Zero-zero is the universal hero".

Night-Before-the-Exam Sheet

Fact sheet

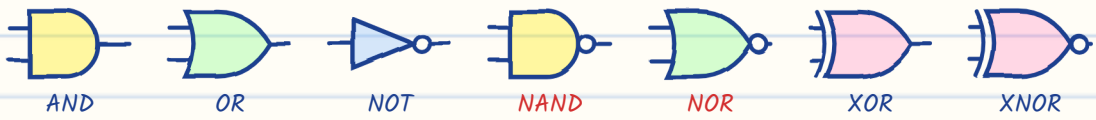
AND: A·B · OR: A+B · NOT: A'
NAND: (AB)' · NOR: (A+B)'
XOR: A'B + AB' (differ → 1)
XNOR: AB + A'B' (same → 1)
(AB)' = A'+B' · (A+B)' = A'B'
A⊕0 = A · A⊕1 = A' · A⊕A = 0
n inputs → 2^n rows · $2^{(2^n)}$ functions
Universal: NAND, NOR only
NAND: NOT 1, AND 2, OR 3, XOR 4
NOR: NOT 1, OR 2, AND 3, XNOR 4
SOP → NAND-NAND · POS → NOR-NOR
TTL: $V_{IL} \le 0.8$ V, $V_{IH} \ge 2$ V, NM 0.4 V
7400 NAND · 7402 NOR · 7404 NOT
7408 AND · 7432 OR · 7486 XOR

→ Top 12 one-liners for MCQs

1. Basic gates: AND, OR, NOT
2. Universal gates: NAND, NOR
3. Only single-input gate: NOT
4. Output 1 for all-0 inputs: NAND, NOR, XNOR
5. Output 0 for all-1 inputs: NAND, NOR, XOR
6. XOR is also called "inequality" gate
7. XNOR is the "equality / equivalence" gate
8. NAND(A, A) = NOR(A, A) = A'
9. Min. NAND for XOR = 4
10. Positive AND = negative OR
11. Half-adder sum = XOR, carry = AND
12. 7404 = hex inverter (6 NOTs)

Top traps OR is 0 **only** for all-0 inputs · NAND/NOR not associative · dual ≠ complement · XOR, XNOR and AND, OR are **not** universal · textbook 5-NAND XOR is not the minimum

All tricks together AND = all, OR = any · "N" = bubble = flip the column · break the bar, change the sign · bubble pushing swaps AND ↔ OR · XOR = different, XNOR = same · 7400 = universal hero



draw all 7 symbols from memory before the exam

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. Which gate gives output 1 only when all inputs are 0?
(a) NAND (b) NOR
(c) XOR (d) OR

2. Minimum number of 2-input NAND gates to build XOR:
(a) 3 (b) 5
(c) 4 (d) 6

3. $A \oplus 1$ equals:
(a) A (b) 1
(c) 0 (d) A'

4. Rows in the truth table of a 4-input gate:
(a) 8 (b) 16
(c) 4 (d) 32

5. IC 7404 contains:
(a) 6 inverters (b) 4 NAND
(c) 4 NOR (d) 4 XOR

6. A positive-logic AND gate works in negative logic as:
(a) NAND (b) NOR
(c) OR (d) AND

7. $(A + B)'$ equals:
(a) $A'B'$ (b) $A' + B'$
(c) AB (d) $(AB)'$

8. Number of distinct Boolean functions of 2 variables:
(a) 4 (b) 8
(c) 2 (d) 16

9. Minimum 2-input NOR gates for an AND gate:
(a) 2 (b) 3
(c) 1 (d) 4

10. In C, 12 ^ 10 evaluates to:
(a) 6 (b) 8
(c) 14 (d) 2

Answer key → 1-b · 2-c · 3-d · 4-b · 5-a · 6-c · 7-a · 8-d · 9-b · 10-a

Written question Implement a 2-input XOR gate using the minimum number of NAND gates only. Draw the circuit and prove it algebraically. 8 marks

1. **Circuit:** G1 = NAND(A, B) → P; G2 = NAND(A, P) → Q; G3 = NAND(B, P) → R; G4 = NAND(Q, R) → Y (diagram on sheet 5). 2
2. $P = (AB)' = A' + B'$ (De Morgan); $Q = (A(A' + B'))' = (AA' + AB')' = (AB')'$; $R = (A'B')'$ 3
3. $Y = (Q \cdot R)' = Q' + R' = AB' + A'B = A \oplus B$ ✓ (De Morgan + double complement) 2
4. Check: AB = 00 → P = 1, Q = 1, R = 1, Y = 0 ✓; AB = 10 → P = 1, Q = 0, Y = 1 ✓. **4 gates = minimum.** 1

→ Before the exam ✓

- ✓ Can draw all 7 symbols + truth tables
- ✓ Can build NOT/AND/OR from NAND and NOR
- ✓ Know minimum gate counts (4 NAND XOR)
- ✓ Can apply De Morgan & bubble pushing
- ✓ Know 7400 … 7486 and V_{CC}/GND pins

Also practise (written):
① XNOR from 4 NOR gates
② Majority voter with NAND only
③ Prove De Morgan by truth table
④ Even-parity generator for 4 bits